

Tilak Patel

Incoming Machine Learning Intern @ SRC
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EDUCATION

Northeastern University

April 2027

Major: B.S. Computer Science (Artificial Intelligence)

Minors: Computer Engineering and Robotics

Relevant Coursework: Object-Oriented Design, Computer Systems, Algorithms, LLM-Integrated Systems, Web Dev

TECHNICAL SKILLS

Programming: Python, Java (Fluent); JavaScript, TypeScript, MongoDB, C/C++, HTML, CSS (Intermediate)

Data/AI: PyTorch, NumPy, Pandas, Scikit-learn, Generative AI, LLMs, NLP, Retrieval Augmented Generation, Multi-Agent Orchestration, Prompt Engineering, Agentic Design, Computer Vision & Classification

Systems: Assembly, OS, VMs, RISC, RTOS, freeRTOS, Threads, Parallelism/Concurrency, Synchronization

Tools/Libraries: AWS, Git, Linux, Docker, Jupyter, Azure, Postman, Gradle, pytest, GitHub, VSCode, Redux, Material UI, Tailwind CSS, Framer Motion, Node.js, Java Swing, OpenAI API, Hugging Face Transformers, Flask, FastAPI

Interests: Star Wars, Pop Culture, Cooking, Indian Classical Music, Sitar, Gaming, Building Gaming PCs, Philosophy

EXPERIENCE

Research & Development

Sept. 2024 – Present

Later Boston, MA

- Conducting R&D at Later.com through the DASH Research Practicum at Northeastern, focusing on AI-driven categorization using CLIP, CNNs, and image segmentation for a 1 million image dataset.
- Exploring text embeddings and models (word2vec, doc2vec) to enhance text-based media classification.
- Leveraging Python and machine learning to develop innovative solutions for image and text data challenges.

Embedded Software Developer

Sept. 2024 – Present

NU Electric Racing (NERO Team)

Boston, MA

- Developing real-time car dashboard GUI with C++, Qt, and QML, displaying speed, temperature, and battery data.
- Implementing MQTT and Web Sockets for efficient sensor data transmission and system debugging.
- Contributing to feature development, including an interactive Flappy Bird game on the dashboard.

National Development Committee

Sept. 2024 – Present

BAPS Swaminarayan Sanstha

North America

- Developing weekly syllabi for hundreds of BAPS centers across the U.S., fostering cultural and spiritual growth.
- Leading national youth events, promoting leadership and community service.
- Collaborating to design large-scale programs engaging thousands of North American Hindu youth.

Generative AI Intern

July 2024 – Sept. 2024

Empowerreg AI

Seattle, WA

- Engineered production-ready visualizations for Risk Annotation Matrix Product, enhancing data interpretation.
- Developed visualizations for multi-dimensional risk and complaint data analysis of medical devices from LLM's.
- Optimized processing of 1,000+ lines of JSON medical device data using FastAPI, reducing transmission time.

PROJECTS

Portfolio Website

July 2024

React, Vite, Framer Motion, Tailwind CSS, Git

- Developed responsive portfolio using React and Vite, achieving optimal performance.
- Implemented Framer Motion animations, increasing average session duration.
- Designed modern UI with Tailwind CSS, enhancing visual appeal and user feedback.

Smart Summarization NLP & ML Tool

July 2024

Python, PyTorch, BERT, Transformers, Google Colab, Seaborn, Matplotlib

- Developed advanced NLP tool using BERT, achieving 6.69% improvement in training loss over 2 epochs.
- Optimized data pipeline, processing 287,113 samples to 50,000, reducing training time to 30-45 minutes on Google Colab's T4 GPU.
- Implemented PyTorch-based model with mixed precision training and gradient accumulation for 4x effective batch size increase.